

Nate Cirino

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Available May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Foundations of Data Science, Systems Security, Cybersecurity, Computer Networks, Computer Systems, Algorithms & Data Structures, Foundations of Software Engineering, Object-Oriented Design, Digital Design & Computer Architecture

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Building a supplier-agnostic e-label API unifying 4 suppliers behind one contract, powering outlet launches
- Shipped real-time POS checkout flows: in-store scan-to-basket switching and live associate-entry customer sync
- Own front-of-house device stack across all Wayfair retail brands: e-labels, mobile, and customer-facing POS

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Architected Python/Flask backend with RAG pipeline on Bedrock Knowledge Bases for personalized workouts
- Evolved RAG stack from LangChain to Vespa to zero-ops Bedrock KB, improving retrieval and reducing overhead
- Sustained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows
- Continue as technical advisor on architecture and agentic-workflow direction

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Integrated Expert AI (GenAI clinical decision support) into CME microservices, reaching 250K+ clinicians
- Optimized Expert AI search and retrieval across 20+ microservices, improving clinical query relevance at scale
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Overhauled Relay JMS message-broker serialization with json-io across 20+ services, eliminating timeouts
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

PROJECTS

Sentinel | *Python, FastAPI, Claude, pgvector, Next.js, OpenTelemetry, Docker*

Apr. 2026 – Present

- Built an LLM agent autonomously triaging incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Gated remediation behind human-in-the-loop approval; running live in SculptAI production triaging incidents

NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Trained CNN models in PyTorch on sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined architecture and loss formulation to address class imbalance and sensor noise, improving detection
- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across multi-node Raft cluster

SKILLS & HONORS

Languages: Python, Java, Rust, TypeScript, JavaScript, C, C++, Groovy, Bash

AI / ML & LLM Sys: PyTorch, TensorFlow, Pandas, RAG (Bedrock KB, LangChain, Vespa, pgvector),
tems: prompt orchestration & caching, agentic tool-use, evaluation loops; Claude Code, Devin, Cursor, GitHub Copilot, CodeRabbit

Backend & Infrastructure: FastAPI, Flask, Java Spring, Node.js, Express, REST/GraphQL APIs, PostgreSQL, JMS message queues, Docker, Kubernetes, AWS, Azure, Terraform, Jenkins, Buildkite, Datadog, OpenTelemetry, Linux, React, Next.js

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor